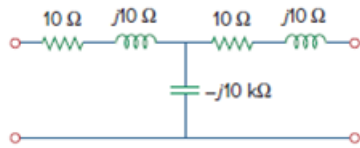


Problem

Find the **ABCD** parameters for the circuit in Fig. 19.100.

Figure 19.100



Step-by-step solution

Step 1 of 9

Refer to Figure 19.100 in the text book.

Connect the voltage source $\mathbf{V}_1$ as shown in Figure 1. Identify the currents and voltages in the circuit.

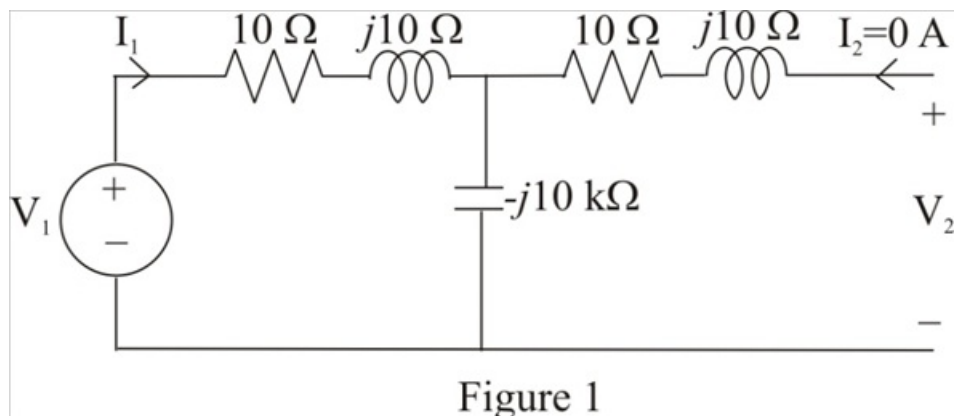


Figure 1

Step 2 of 9

Consider the expression for the voltage $\mathbf{V}_1$.

$$\begin{aligned}\mathbf{V}_1 &= \mathbf{I}_1(10 + j10 - j10 \text{ k}) \\ &= \mathbf{I}_1(10 - j9990)\end{aligned}$$

Therefore, the expression for the voltage $\mathbf{V}_1$ is $\mathbf{I}_1(10 - j9990)$.

Consider the expression for the voltage $\mathbf{V}_2$.

$$\mathbf{V}_2 = \mathbf{I}_1(-j10 \text{ k})$$

Therefore, the expression for the voltage $\mathbf{V}_2$ is $\mathbf{I}_1(-j10 \text{ k})$.

Step 3 of 9

Determine the expression for the open circuit voltage ratio A .

$$A = \frac{V_1}{V_2}$$

Substitute $I_1(10 - j9990)$ for V_1 and $I_1(-j10 \text{ k})$ for V_2 .

$$\begin{aligned} A &= \frac{I_1(10 - j9990)}{I_1(-j10 \text{ k})} \\ &= 1 - j0 \\ &= 1 \end{aligned}$$

Therefore, the expression for the open circuit voltage ratio A is $\boxed{1}$.

Step 4 of 9

Determine the value of the open circuit transfer admittance C .

$$C = \frac{I_1}{V_2} \text{ Substitute } I_1(-j10 \text{ k}) \text{ for } V_2.$$

$$\begin{aligned} C &= \frac{I_1}{I_1(-j10 \text{ k})} \\ &= j100 \mu\text{S} \end{aligned}$$

Therefore, the value of the open circuit transfer admittance C is $\boxed{j100 \mu\text{S}}$.

Step 5 of 9

Connect the voltage source V_1 and short the port 2 terminals as shown in Figure 2. Identify the currents and voltages in the circuit.

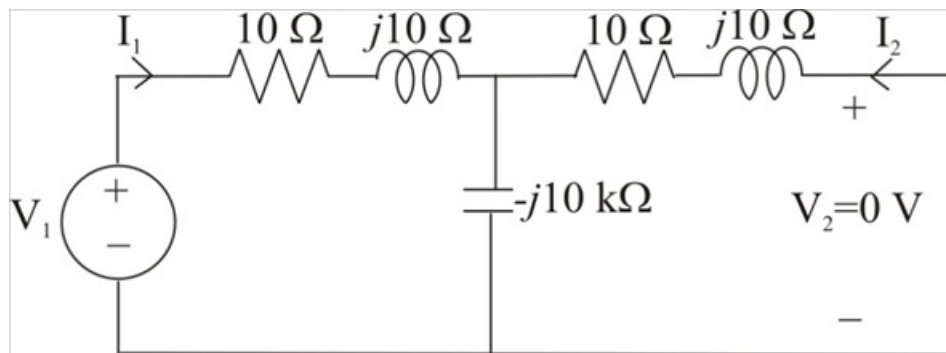


Figure 2

Step 6 of 9

Consider the expression for the voltage V_1 .

$$\begin{aligned} V_1 &= I_1 \left(10 + j10 + \left[(-j10 \text{ k}) \parallel (10 + j10) \right] \right) \\ &= I_1 \left(10 + j10 + \frac{(-j10 \text{ k})(10 + j10)}{-j10 \text{ k} + 10 + j10} \right) \\ &= I_1 \left(10 + j10 + \frac{100000 - j100000}{10 - j9990} \right) \\ &= I_1 (10 + j10 + 10 + j10) \\ V_1 &= I_1 (20 + j20) \end{aligned}$$

Therefore, the expression for the voltage V_1 is $I_1 (20 + j20)$.

Step 7 of 9

Consider the expression for the current I_2 .

$$\begin{aligned} I_2 &= -I_1 \left(\frac{-j10 \text{ k}}{10 + j10 - j10 \text{ k}} \right) \\ &= -I_1 \left(\frac{-j10 \text{ k}}{10 - j9990} \right) \\ &= -I_1 (1 - j0) \\ &= (-1) I_1 \end{aligned}$$

Therefore, the expression for the current I_2 is $I_1 (-1)$.

Step 8 of 9

Determine the expression for the negative short circuit transfer impedance B .

$$B = -\frac{V_1}{I_2}$$

Substitute $I_1 (-1)$ for I_2 and $I_1 (20 + j20)$ for V_1 .

$$\begin{aligned} B &= -\frac{I_1 (20 + j20)}{I_1 (-1)} \\ &= (20 + j20) \Omega \end{aligned}$$

Therefore, the expression for the negative short circuit transfer impedance B is

$$\boxed{I_1 (20 + j20)}.$$

Step 9 of 9

Determine the value of the negative short circuit current gain D .

$$D = -\frac{I_1}{I_2}$$

Substitute $I_1 (-1)$ for I_2

$$\begin{aligned} D &= -\frac{I_1}{I_1 (-1)} \\ &= 1 \end{aligned}$$

Therefore, the value of the negative short circuit current gain D is $\boxed{1}$.